

Types of Orchestration



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Storyline



Quinn
Systems Administrator

Problem

- What type of orchestration to use?

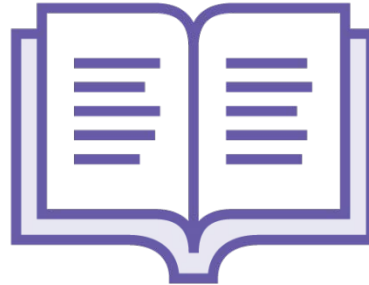
Questions

- What types are there?
- How do they work?

Types of Orchestration



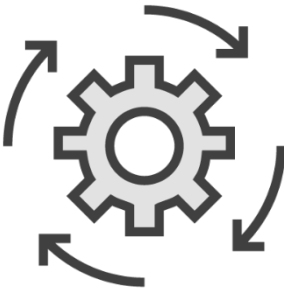
Operating system task
schedulers



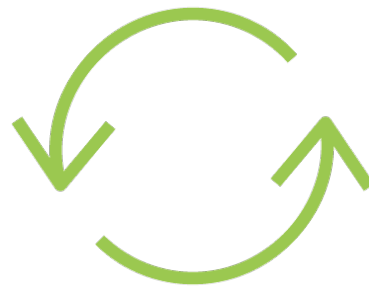
Runbook orchestration



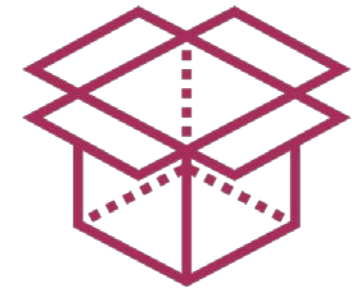
Cloud-based
automation



Configuration
management



Continuous
integration

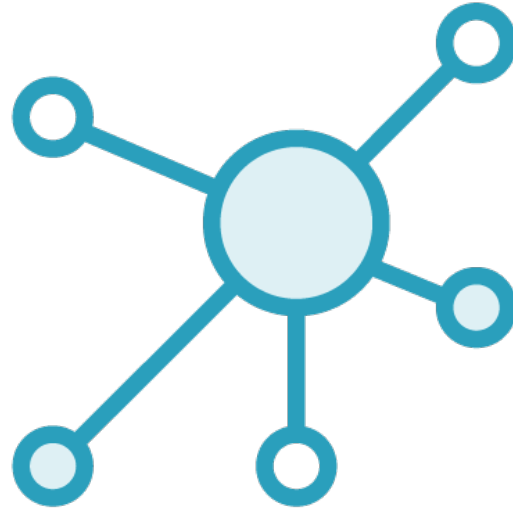


Continuous
deployment

Orchestration Communication Methods



Locally



Remoting
protocol



Agent



Service

Operating System Task Scheduler

Is a built in service that allows you to perform automated tasks on a chosen computer



Operating System Task Schedulers



Triggers

- Scheduled time
 - Hourly, daily, weekly, monthly
- Events

Actions

- Run a specified program, script, or automation

Communication method

- Local

Runbook Orchestration

A workflow management system that allows you to coordinate, deploy, and monitor automation



Runbook Orchestration



Requires

- Management server
- Runbook runners
- Database

Web portal or graphical interface

- Creating, edit, monitoring and auditing workflows and runbooks

Communication method

- Agent

Cloud-based Automation

Provides end-to-end automation workflows that coordinate lower-level automation through an API and/or a web portal for cloud resources



Cloud-based Automation



Provided as a service

- API
- Web portal

Manages and coordinates

- Cloud runbooks
- Cloud workflows
- Lower-level automation tasks
- Cloud resources

Communication method

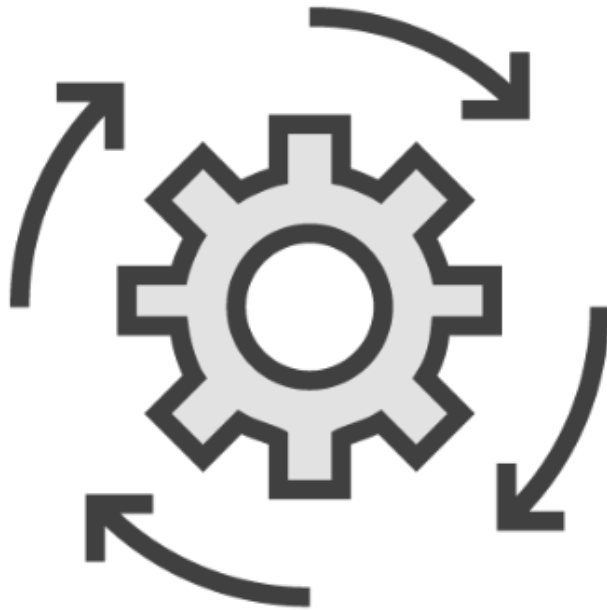
- Cloud service

Configuration Management Tools

Provides the ability to manage system state changes and provisioning of infrastructure through automation



Configuration Management Tools



Requires

- Central server
- Agents or remote protocol

Declarative infrastructure

- DSL (Domain Specific Language)

Communication method

- Agent or remoting protocol

Continuous Integration Tools

Automates software development build processes that are triggered by code changes to central code repositories



Continuous Integration Tools



Requires

- Continuous integration server
- Database
- Build agents

Provides automation

- Software build process
- Testing

Communication method

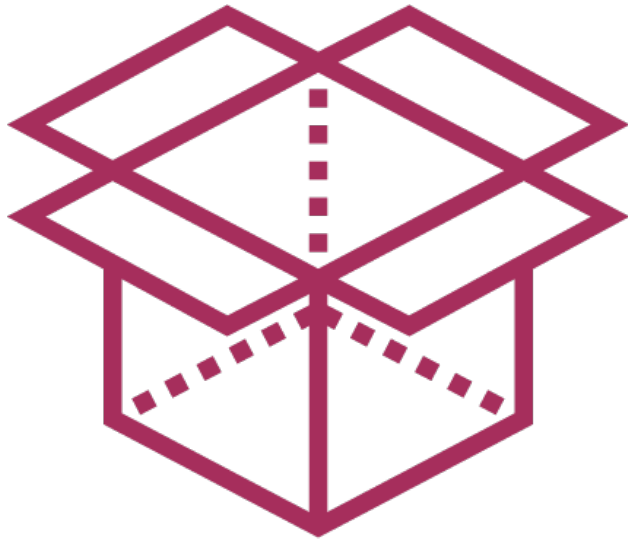
- Agent

Continuous Deployment Tools

Are designed for automating the deployment of software releases



Continuous Deployment Tools



Requires

- Continuous deployment server(s)
- Database
- Agents

Provides automation

- Deployment of software across different environments

Communication method

- Agent

Summary



Types of orchestration

- Built-in, runbook platforms, cloud-based, pipeline tools

How the different types of orchestration work

Orchestration communication methods

- Local
- Remoting protocol
- Agent
- Cloud service

